

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech Mid Semester Exam

Semester: 6th**Subject Code: 303105311****Subject Name: Quant & Reasoning****Date: 27/01/2025****Time: (1hr: 30min)****Total Marks: 40**

Sr. No.		Marks	Co/Po	Blooms Taxonomy
Q.1	(A) Five One line Questions (1) Find the HCF (150, 225, 300). (2) If the number 97215A6 is divisible by 11, find the smallest whole number in the place of A. (3) Find the Least number which divided by 12, 16 & 18 will leaves in each case a Reminder is 5. (4) Whose 20% is 280? (5) A Person travels 120 miles in 2 hours, 150 miles in 3 hours and 180 miles in 4 hours. What is the person's average speed (in miles per hour)?	05	2 2 2 2 1	Apply Apply Apply Apply Evaluate
	(B) Five Fill in the blanks (1) The Sum of the Infinite series $S_{\infty} = \underline{\hspace{2cm}}$. (2) Find the value of $\binom{15}{4} = \underline{\hspace{2cm}}$. (3) If a man stands facing the North at the time of sunset then his shadow will be towards his _____. (4) In a simultaneous toss of 3 coins, then a probability of 2 heads and 1 tail are together is _____. (5) If $5:3 = 15:x$ Then $x = \underline{\hspace{2cm}}$.	05	2 2 1 1 2	Apply Apply Evaluate Evaluate Apply
Q.2	Attempt any four(Short Questions)	12		
	(1) Two dice are thrown together. What is the probability that the sum of the number on the two faces is divisible by 4?		3	Analyze
	(2) How many ways can the letters of the word "ORANGE" be arranged, in such a way that O and A are never together?		1	Evaluate
	(3) Rudra Starts from Point A and drives 13 km towards East. Then He takes a right turn and drives 15 km, turns right and drives 17 km, He then takes a right turn and drives 16 km, He takes a final right turn and drives 4 km and stop at a Point P. How far (shortest distance) and towards Which direction should he drive in order to reach Point A again?		1	Evaluate
	(4) A sum of money is to be distributed among A, B, C, D in the proportion of 5 : 2 : 4 : 3. If C gets Rs. 1000 more than D, what is B's share?		2	Apply
	(5) The Sum of First 8 terms of A.P is 56 and the Sum of First		2	Apply

	20 terms is 260. Find the first term (a) and common difference (d).			
Q.3	Attempt any two questions	08		
	(1) In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there is Rs. 30 in all, then find the number of coins of each type respectively.		1	Evaluate
	(2) Ratio of age's of A & B, 6 years ago was 3 : 4. C is 12 years elder to A. Sum of the Present age of B & C is 80 years. Find the difference of B & C age's 5 years later ?		2	Apply
	(3) How many different words can be formed using all the letters of the word " ALLAHABAD " when both L Do not occur together?		3	Analyze
Q.4	(A) In a school having roll strength 286, the ratio of boys and girls is 8 : 5. If 22 more girls get admitted into the school, the ratio of boys and girls becomes?	05	2	Apply
	(B) Eight People - A, B, C, D, E, F, G and H are sitting around a circular table facing the centre, not necessarily in the same order. Three people are sitting between A and D. B is sitting second to the right of A. C is to the immediate right of F. D is not an immediate neighbour of either F or E. H is not an immediate neighbour of B. (I) What is E's position with respect to G ? (II) How many people are sitting between H and A When counted from the right side of H ?	05	1	Evaluate
	OR			
	(B) There are five different buildings A, B, C, D & E. The following are the clues given: A is to the right of B. E is to the left of C and right of A. B is to the right of D. How many buildings are there between D and C?	05	1	Evaluate